

[Home](#) > [My Courses](#) > [Portfolio Management](#) >[M4: Behavioral Finance and Applications to Portfolio Theory](#) > Lesson Notes

Lesson Notes

MODULE 4 | LESSON 3

BEHAVIORAL PORTFOLIO THEORY

Reading Time	60 minutes
Prior Knowledge	Lesson 1 and Lesson 2 of Module 4.
Keywords	Behavioral Portfolio Theory (BPT), Adaptive Market Hypothesis (AMH), relative wealth, standard of living risk, Behaviorally Modified Asset Allocation (BMAA)

The previous lesson dealt with the formal aspects of prospect theory and how it can be applied to real-life situations. In particular, we investigated how prospect theory is capable of explaining certain puzzles and anomalies that are difficult or impossible to justify within the realm of traditional finance.

In this lesson, we look at portfolio theory. Traditional finance with its formulas and perfectly rational individuals is just too difficult for many individuals to accept, and this is true in particular when investors' portfolios suffer losses. Behavioral portfolio theory and behaviorally modified asset allocation are the tools proposed by behaviorists to better serve investors and ultimately assist them in reaching their financial goals.

1. Problems with Traditional Portfolio Theory

Up to this point, we have learned more than a little bit about how to construct optimal portfolios. However, all we did was based on requiring that individuals act as **rational economic men (REM)** who are able to access all relevant information and capable of using new information to upgrade their solutions. We also insisted on them being risk-averse. The first two lessons of this module, on the other hand, argued that human beings do not exactly behave like the theory tells us they should.

TF makes one key assumption: **Markets are efficient** and prices reflect the fundamental value of assets. Furthermore, TF endorses the notion that, as soon as new information becomes available, it is quickly and correctly transferred to market prices. The implication of these two postulates is that managers can focus on identifying which portfolios on the efficient frontier are compatible with their clients' objectives in terms of risk and return. Nothing else seems to matter. We could discuss whether markets are truly strong efficient or semi-strong efficient, or even just weak efficient.

However, when prices no longer represent the correct intrinsic value of assets, the traditional approach to portfolio theory is flawed and not just in a minor way.

We must also notice one further aspect of TF: It assumes implicitly that investors, in going through different stages in their lives, are able to save and invest in the early stages to fund their expenses in retirement. While on paper this seems a sensible behavior to expect from rational investors, it requires that investors are capable of self-control, i.e., they are capable of delaying short-term spending and gratification in exchange for achieving long-term goals. Unfortunately, there have been plenty of experiments that display that actual individual behaviors run contrary to self-control. If that were not enough already, in many situations, even the very process of thinking in terms of saving now to consume later represents a challenge since individuals are prone to suffer from biases like mental accounting and framing. For instance, it has been observed that individuals treat the money they earn as wages differently from the money they earn as bonuses or by winning a lottery: If they perceive a bonus as part of their regular compensation, they are more likely to spend it while if they perceive it as exceptional instead, they are less likely to spend it and may save a bigger percentage of it.

Consider again the equity premium puzzle discussed in the previous lesson. TF is quite clear in that the value asset is calculated by considering the expected cash flows over the life of the investments and discounting them using a rate that depends on the risk-free rate and some suitable risk premium. This requires that one knows the correct discount rate to use. BF, on the other hand, adds a *sentiment premium* to the discount factor. Roughly speaking, this sentiment premium can usually be estimated by considering the dispersion of the analysts' forecasts where a high dispersion implies a high sentiment premium (more dispersion = more risk). In TF, there is no need for a sentiment premium, and besides, if it were added and predictable, it would be, or at least it could be, exploited and eventually eliminated, possibly via arbitrage. The presence of a sentiment risk premium can change the perception of risk and reward among individuals, and it raises the question of whether sentiment can be considered a priced source of risk or if it causes mispricing instead. (Readers interested on these issues should read the paper by Shefrin and Statman listed in the References section).

These are serious issues that should be addressed.

2. Behavioral Portfolio Theory (BPT)

At the risk, perhaps, of repeating ourselves, though an old saying in Latin insists that *repetita iuvant*, i.e., to repeat is beneficial, we recall that in TF, we end up with a diversified portfolio that satisfies the investor's preference for risk and return by selecting possible combinations of assets from the set of mean-variance efficient portfolios (the efficient frontier). That is the sensible thing to do as TF is built upon the tenet that investors are concerned about expected return and risk of the whole portfolio, i.e., a combination of assets that is considered in its totality.

Optimal portfolios in TF can be regarded as the optimal portfolios of CAPM investors, and as we saw in Module 1, they combine a market portfolio with a risk-free asset.

Optimal portfolios of BPT investors resemble combinations of bonds and lottery tickets instead. This happens because BPT is based on a very different principle: Investors build their portfolios in layers, and for each layer, there is a different risk and return preference. Hence, investors do not compute an overall optimal portfolio, but an optimal portfolio for each layer. This is well explained in

the 2000 paper by Shefrin and Statman, a reading that is not very technical and strongly recommended to students who are interested about the behavioral perspective.

According to BPT, portfolios are built to maximize wealth with one important constraint: Given an investor's most important goals, we want a low probability of failing to meet these objectives. Assets are separated in layers based on risk. The low-risk layers will contain bonds or riskless investments; the high-risk level will consist of certain speculative equities, private equity investments, derivatives, etc. This version is called a **multiple mental account version (BPT-MA)** by Shefrin and Statman. There exists also a single mental account BPT version (**BPT-SA**) in which investors, like mean-variance investors, integrate their portfolios into a single mental account by considering covariance among the different layers. However, this version is less popular, and when we refer to BPT, we will do so meaning BPT-MA.

Each investor has certain goals in mind: Meeting basic needs, paying for children's schooling or college in the future, providing for a sick parent in their old age, buying a fancy car, buying a vacation home, etc. Some of these goals will be deemed essential; others can be considered just desirable. For instance, providing for basic needs or for a sick member of the family would be considered essential, and therefore, the investor has little or no desire for speculative investments as keeping the risk low is the priority. Buying a vacation home in Aspen, Colorado, may be very desirable, but it is not essential. In this case, the investors may be willing to accept riskier investments to get closer to this goal but would not produce dramatic consequences if the required return is not achieved.

With all this in mind, the process of constructing portfolios according to the behaviorists' perspective goes through the following steps:

- Each layer's fund allocation is determined by the investor's goals. If the goal requires a high return, funds will be allocated to a high return/high risk layer. On the other hand, if the goals require a low risk, then financial resources will be allocated to the low return/low risk layer.
- Asset selection will be done on a layer by layer basis.
- In each layer, the number of assets will reflect the investor's risk aversion. In practical terms this means that if an investor is risk averse (concave utility curve), they will hold a more diversified portfolio involving a larger number of assets than an investor whose utility curve is convex or concave and convex. This is applied on a layer by layer basis.
- In the case an investor comes across some information that is not public knowledge, they can tweak the portfolio composition toward more concentrated positions.
- When investors are loss-averse, they will hold larger cash positions to avoid liquidity requirements forcing them to sell assets at a loss. In addition, there could be instances where an investor holds on to certain assets even if their potential performance is not attractive simply because they want to avoid realizing losses.

The optimal portfolio that comes out of this process could give the impression of being diversified, but since the layers were constructed without considering the correlation among the different layers, it is likely a sub-optimal combination. The reason BPT-MA is relevant is that it is able to explain what one observes in real life:

- individuals holding both low risk assets and high risk assets
- holding excess cash in low risk layers and excessive risky assets in high risk layers and only modest amounts of moderately risky assets.

3. From Efficient Market Hypothesis to Adaptive Market Hypothesis

The concept of the adaptive market hypothesis (AMH) originated in a 2004 paper by Andrew Lo and can be summarized with the formula:

$$\text{AMH} = \text{EMH} + \text{bounded rationality} + \text{satisficing} + \text{"evolution"}.$$

In essence, AMH assumes that market participants, to be successful, apply heuristics instead of hard core optimization. In addition, they do so as long as these heuristics work, and once they stop working, participants modify them and/or try to come up with new ones.

This behavior suggests that **winning strategies are an evolutionary game**. Nevertheless, there is not just a change in how the "game" is played, but a change in the goals as well: Unlike TF where the goal is maximizing expected utility, now the objective is to survive in competitive markets, and those who cannot adapt may not be able to survive.

The presence of satisfice should be recognizable from the fact that in the previous paragraph we acknowledged the fact that market participants are not performing a hard core optimization, but based on the amount of information that they consider sufficient, they make decisions to reach sub-goals on their path towards achieving their ultimate goal. The use of heuristics means that economic agents are able to come up with decisions via trial and error, and these decisions, unlike theorems, are not supposed to work all the times. Nevertheless, investors keep using and updating them until, at a certain point, their rules are reflected in market pricing and no longer work.

AMH produces several practical consequences:

1. The relationship between risk and return is not stable in response to competitive changes;
2. Active management makes sense since it is possible to find arbitrage opportunities to exploit;
3. Strategies do not work all the time;
4. To be successful and stay successful, one has to adapt and innovate, and those who are best at changing and innovating will survive. The others may not.

We could ask ourselves one question: why are many investors ultimately not able to construct portfolios that are optimal in the sense of TF? Arguably, the most fitting answer is that they do not understand them and when investors do not understand them, they are not likely to follow TF's principles in the long run. What if they gave the money to a portfolio manager who is skilled and understands TF? That could be a solution; however, the experience in the field is that these investors would be happy as long as their portfolios do well but are likely to complain or even withdraw their money from the fund when things do not go the way they envisioned. Consequently, they are more than willing to blame the portfolio manager (or themselves for listening to their advisors) for any unfavorable outcome. The self-control bias can be applied even to how investors behave towards portfolio managers because, at a personal level or in the relationship with a portfolio manager, they could lack the necessary discipline to be TF investors and follow a path they do not understand or, at least, do not fully understand.

4. Behaviorally Modified Asset Allocation

Behaviorally modified asset allocation, (BMAA) is an approach to asset allocation in which clients' behavioral biases are taken into consideration and incorporated into the allocations themselves.

What is the advantage of doing so? It serves to help financial advisors or portfolio managers to build portfolios for their clients in such a way that clients will not stray from them when they encounter periods during which their portfolios, for whatever reason, are under-performing compared to their expectations.

The last point is very important: If a client is upset by the fact that their portfolio lost, say, 10% over a period of a few months, they may abandon an otherwise sound strategy. Worse still, they may do so at a low point and, to add insult to injury, do it right before their portfolio begins to recover. Of course, this assumes that the poor outcome is only temporary.

For financial advisors, it is hard to advise their clients if their clients are not listening. BMAA is meant to help the relationship between financial advisor and client.

The ultimate goal of BMAA is to determine a strategy in advance that a client is more likely to stick to even during adverse periods.

A BMAA is created in several steps:

1. determine a strategic asset allocation that is optimal and consistent with TF;
2. evaluate the client's biases and, possibly, determine if they are mainly cognitive or emotional.
3. take into consideration the level of wealth of the client versus their lifestyle and needs/desires.

The first two steps are self-explanatory and do not require any further comments. The third step is perhaps less evident, and it can be better understood in terms of determining the size of the **standard of living risk**.

To illustrate this concept, considers two clients, both living a fairly lavish lifestyle, and let's call them Client A and Client B. Now:

- Client A has a financial portfolio worth \$10 million, an annual salary of \$500,000, and no children or other family members who require his assistance.
- Client B instead has a financial portfolio of \$1 million, an annual salary of \$250,000, and two children who are about to go to college for which he would like to pay. Additionally, he is taking care of his brother who suffers from a severe illness, is unable to work, and needs frequent expensive medical care that costs about \$100,000 a year.

While they are both reasonably wealthy, in terms of relative wealth, Client A is in a much better position to maintain their basic lifestyle, and in particular, there is a very low risk that he might be unable to fulfill his essential needs. On the contrary, Client B is almost "living on edge."

Now, the concept of essential needs is not synonymous with "just enough to survive." Some individuals may think of lavish lifestyles as necessary to enjoy life at its fullest. Other individuals, on the opposite side of the scale, could think of essential needs as just putting food on the table and a roof over their heads. In some cases, it could mean being able to fulfill basic needs and being able to care for a sick member of the family.

What we are stressing here is that the definition of essential needs is personal to the particular client. A financial advisor will have to be able to recognize which is what.

The point is that Client A has a low standard of living risk and could navigate troubled markets without much risk of being forced to give up his lifestyle. On the other hand, Client B has a high standard of living risk since even a moderate economic downturn could jeopardize his ability to meet his basic needs and obligations.

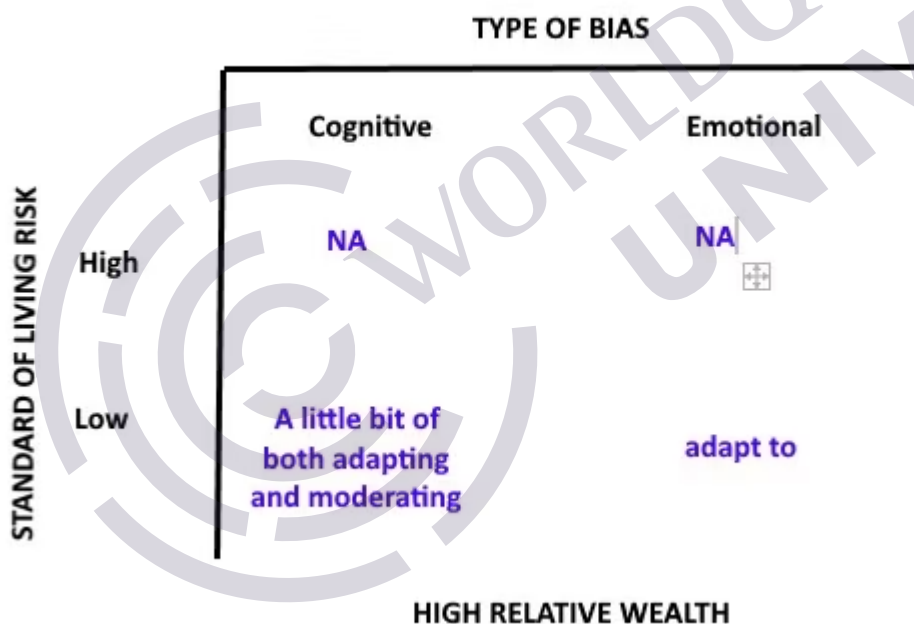
Having stressed this point, to come up with a BMAA portfolio, three initial determinations are required:

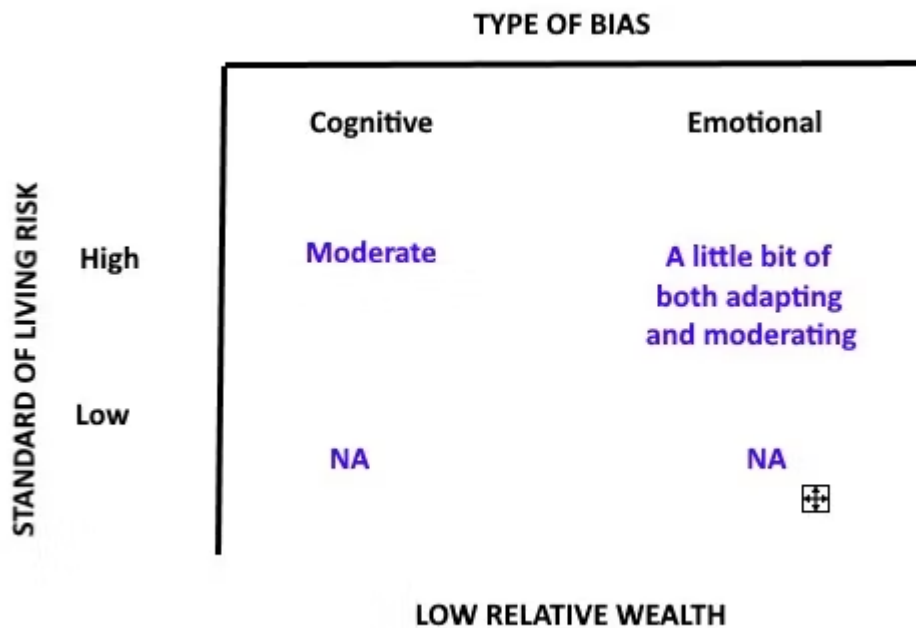
1. The level of relative wealth vs. financial goals. To facilitate the process, financial goals can be divided in three main groups: (a) basic needs (food, rent or mortgage, taking care of family, a specific lifestyle that is considered very important for the person), (b) priorities and desires (donating to charity, establishing endowments), (c) aspirational (owning an exotic car, a second home on the French Riviera, or even something much less flashy as leaving a bequest at death);
2. The client's level of standard of living risk (high or low);
3. The types of biases that the client displays (cognitive or emotional).

Then, we have to make a fourth and last determination:

1. Decide whether to mitigate/moderate or adapt to the client's biases. This is done according to a three-dimensional matrix (see figure 1):

Figure 1: BMAA at Work





Let's try to understand the process of how the entries in the tables are chosen.

A high level of wealth vs. essential needs means a low standard of living risk. On the other hand, a low relative wealth in general means a high standard of living risk.

We have already explained that cognitive biases are easier to correct or at least moderate because it is possible to correct the faulty reasoning associated with them. On the other hand, emotional biases are more difficult to correct as the faults are deeply embedded in an individual's psyche. In the case of wealthy individuals with a low standard of living risk and emotional biases, we can indulge the client and adapt (within reason!) to their choices because they can afford to deviate more from an optimal portfolio than less affluent individuals. If we were dealing with a low relative wealth client with a high standard of living risk, regardless of whether the behavioral biases are emotional or cognitive, there is a need to introduce at least some moderation so that the client's portfolio does not stray too much from the optimal one.

What does it mean to adapt and to moderate in practical terms? There is no "one size fits all" sort of rule. However, we could think of something like this table:

Standard of living risk	Primary type of bias	Deviation (up or down) from optimal portfolio weights
Low	Emotional	15% - 20%
Low	Cognitive	5% - 8%
High	Emotional	5% -10%
High	Cognitive	< 5%

Note that the percent deviations listed in the table are just meant to favor a visualization of the relative importance of moderating but are otherwise arbitrary.

For a discussion about these issues, interested students should read the paper by Pompian and Longo listed among the References for this lesson.

Now, let's see where some of these ideas are put to work on an example that is similar to those encountered in real life—or, maybe, a bit dramatized for effect.

Example. You just started working for an important mutual fund in New York City, and John Gwinner is your first client. Following company guidelines, you conduct an extensive interview with Mr. Gwinner and have determined that:

- Mr. Gwinner has a portfolio of \$1,000,000 and a salary of \$180,000 a year, and he is expected to inherit a large sum of money from his mother's estate that he estimates at about \$2,000,000. The company he works for does not provide a defined benefit pension plan.
- Mr. Gwinner plans to retire within the next three years.
- Mr. Gwinner explains that his main goal is to be able to use his financial portfolio and live a modest life after he retires and estimates that, to do that, he will need an income of approximately \$75,000-85,000 a year.
- During the interview you also noticed that Mr. Gwinner displays clear tendencies toward conservatism, framing, representativeness, and endowment biases.

Should you adapt or moderate his portfolio?

Considering Mr. Gwinner's goals and current financial situation, his standard of living risk is low. The biases that were identified are mainly cognitive (the only exception being endowment), and thus, it is easier to moderate his biases. Therefore, according to the table above (which is fictional but useful for an example), we can accommodate a 5% to 8% deviation of his portfolio weights from the optimal values.

Three years later, you meet again with Mr. Gwinner for the annual interview with your clients. Several things have happened to him. His mother became severely ill and needs assisted care 24 hours a day. His son was in a bad car accident and is now unable to work. Mr. Gwinner would like to provide for his niece to attend a prestigious prep school as Mr. Gwinner's son planned. Furthermore, Mr. Gwinner's portfolio did not perform particularly well and is currently valued at \$750,000. He is about to retire and his social security pension will be about \$2,500 a month. His mother's estate now stands at about \$500,000 due mainly to her health situation and medical bills. Mr. Gwinner is concerned that he would now need \$150,000 a year, double the amount he envisioned three years ago. His behavioral biases have not changed.

How does Mr. Gwinner's new situation change your recommendations regarding his portfolio?

Mr. Gwinner is not in dire straits, but his mother's situation, his imminent retirement and consequent loss of income, together with his desire to pay for his niece's prestigious school, have changed his situation quite a bit, and there is a higher standard of living risk than 3 years ago (his needs are realistically felt as essential). This fact combined with his mostly cognitive biases would indicate that you should try to moderate his biases and allow only small deviations from the optimal weights for the assets in his portfolio (< 5%).

In the next lesson, we will cover a couple of formal behavioral models (the BB&K 5-way model and the Pompian model more in detail).

At this point, we may ask ourselves: Where did the fancy portfolio math of the previous modules go?

Well, it is still there in the sense that one still needs to determine an optimal portfolio based on the client's preferences for risk/return. However, we have become amenable to accepting certain deviations from those optimal weights, and in some instances, we can be more accommodating than in others. Nevertheless, we are no longer telling clients that they must invest following nearly definitive rules.

Again, it is not possible to transform an individual who exhibits bounded rationality and behavioral biases into an REM. Doing so can make the client hostile and cause the individual to "rebel" against the portfolio manager's advice or even withdraw the money invested and bring it somewhere else.

5. Conclusion

This lesson explained how to build a financial portfolio according to behavioral finance. It covers the changes that are required to approach the problem with a behaviorist's point of view and introduces the concept of behaviorally modified asset allocations while providing a description of the key parts of it.

The next lesson will look at two well-known models to classify investors in order to assist financial advisors in understanding the personality of their clients in order to better serve them. We will also look at some important sub-optimal behaviors that affect investors and financial advisors/portfolio managers.

References

- Lo, Andrew. "The Adaptive Markets Hypothesis." *Journal of Portfolio Management*, vol. 30, no. 5, 2004, pp. 15-29.
- Pompian, Michael M., and John M. Longo. "Incorporating Behavioral Finance in Your Practice." *Journal of Financial Planning*, vol. 18, no. 3, 2005, pp. 58-63.
- Shefrin, Hersh, and Meir Statman. "Behavioral Capital Asset Pricing Theory." *The Journal of Financial and Quantitative Analysis*, vol. 29, no. 3, 1994, pp. 323-349.
- Shefrin, Hersh, and Meir Statman. "Behavioral Portfolio Theory." *The Journal of Financial and Quantitative Analysis*, vol. 35, no. 2, 2000, pp. 127-151